

TRI-COUNTY AP, NC, USA

WMO: 723079

Lat: 36.298N

Lon: 77.171W

Elev: 68

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93796

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	23.2	-2.0	5.0	31.5	3.8	6.7	31.8	17.1	48.1	14.6	45.5	2.0	350	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	94.8	76.6	91.3	76.4	89.9	75.9	80.2	89.4	78.7	87.9	77.5	86.5	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	143.3	85.1	76.1	136.8	83.6	74.9	131.1	82.1	43.7	89.5	42.3	87.7	40.9	86.7	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	11.9	10.4	10.9	95.6	6.9	4.1	6.0	98.5	2.0	100.9	-1.9	103.2	-6.9	106.1
			9.7	82.2	6.2	1.9	5.2	83.6	1.6	84.7	-1.9	85.8	-6.4	87.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	41.1	44.1	50.2	59.8	67.5	75.2	78.6	77.6	71.8	61.2	50.8	44.7		
	DBStd	15.57	11.27	9.28	9.72	8.67	7.17	5.76	4.72	4.91	5.95	8.60	9.08	9.80		
	HDD50	1008	324	211	123	15	0	0	0	0	0	10	98	227		
	HDD65	3322	744	586	466	204	59	3	0	0	16	179	432	633		
	CDD50	4765	47	47	129	307	542	756	885	856	655	356	123	62		
	CDD65	1604	2	2	7	46	136	309	420	391	221	60	7	3		
	CDH74	13827	8	13	80	452	1184	2789	3967	3372	1501	407	41	13		
	CDH80	5413	0	0	8	115	383	1154	1751	1426	476	98	1	1		
(14) Wind	WSAvg	3.2	3.3	3.6	4.2	4.2	3.4	3.0	2.6	2.6	3.2	2.9	3.1	3.0		
(15) Precipitation	PrecAvg	46.8	3.4	2.8	3.9	3.3	3.7	5.0	5.5	4.8	5.0	3.1	2.8	3.5		
	PrecMax	61.5	6.5	5.7	7.4	6.5	6.5	10.8	9.9	10.9	22.4	6.4	7.5	6.9		
	PrecMin	35.3	1.1	0.5	0.9	1.3	1.6	2.5	2.4	1.3	0.5	0.0	0.3	1.4		
	PrecStd	6.9	1.5	1.4	1.5	1.3	1.3	2.4	1.9	2.5	4.7	1.8	1.8	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	75.5	81.1	88.4	90.8	96.8	97.4	97.5	92.6	88.4	79.1	75.3		
		MCWB	64.8	62.6	63.8	69.9	74.8	75.4	76.1	78.2	73.8	73.5	66.3	67.8		
	2%	DB	70.3	71.2	75.4	83.2	88.1	92.7	94.9	93.5	88.3	82.0	73.5	71.2		
		MCWB	62.6	61.5	62.2	66.8	72.4	75.3	77.1	76.6	73.6	69.5	64.2	64.1		
	5%	DB	64.2	66.3	72.3	79.2	84.3	90.0	91.1	90.4	85.1	78.8	71.7	66.0		
		MCWB	57.8	58.2	59.6	64.4	70.6	75.0	77.3	76.8	73.8	67.8	63.4	61.5		
	10%	DB	58.8	61.8	68.1	75.1	81.0	87.5	89.8	88.2	82.2	74.8	67.1	61.6		
		MCWB	52.6	52.9	58.6	63.4	68.8	74.9	76.7	76.4	71.7	67.6	60.7	56.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	66.5	69.2	74.0	77.2	80.2	82.1	82.8	78.7	76.7	70.7	69.5		
		MCDB	71.7	72.5	74.2	83.2	87.7	90.2	90.8	92.5	85.4	83.1	74.2	74.2		
	2%	WB	63.8	63.5	65.7	70.3	74.6	78.2	80.2	80.4	76.9	74.0	67.1	65.6		
		MCDB	68.9	68.6	72.1	77.3	83.1	88.8	89.4	89.4	83.5	79.0	71.8	69.6		
	5%	WB	59.3	60.0	63.0	67.9	72.8	76.9	78.9	78.6	75.5	71.6	64.5	62.1		
		MCDB	64.0	65.9	69.0	75.2	80.6	86.5	88.3	87.1	81.6	75.9	68.8	65.1		
	10%	WB	54.0	54.4	60.0	65.7	70.8	75.4	77.6	77.2	74.0	68.9	61.6	57.1		
		MCDB	57.3	59.7	66.5	73.1	78.2	84.1	87.1	85.1	79.5	73.4	67.1	61.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	21.0	22.1	23.0	20.7	19.2	18.0	17.8	18.3	21.2	22.3	19.7		
		MCDBR	24.5	27.3	27.0	26.1	23.8	21.9	20.7	20.7	21.2	22.5	24.2	23.4		
	5% WB	MCWBR	18.3	19.1	15.9	12.9	11.5	9.9	9.3	9.3	10.2	12.0	15.7	18.2		
		MCWBR	22.8	25.0	22.5	22.3	20.7	19.3	18.8	18.2	17.2	17.5	19.8	20.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.304	0.319	0.349	0.390	0.438	0.492	0.519	0.513	0.406	0.357	0.322	0.307		
		taud	2.527	2.481	2.429	2.322	2.218	2.115	2.054	2.059	2.356	2.479	2.530	2.553		
	Ebn at Noon	Ebn at Noon	285	292	292	283	270	254	246	245	269	275	275	276		
		Edh at Noon	25	30	34	40	45	50	53	51	36	30	25	23		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	784	1012	1329	1694	1886	2017	1915	1731	1417	1143	870	692		
	RadStd	RadStd	60	115	102	128	147	123	96	99	115	141	69	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+191

Nomenclature: See separate page